

Assignment II [Array & String]

Q1. Longest Common Prefix

Write a function to find the longest common prefix string amongst an array of strings. If there is no common prefix, return an empty string "".

Example 1:

Input: strs = ["flower", "flow", "flight"]

Output: "fl"

Example 2:

Input: strs = ["dog", "race", "car"]

Output: ""

Explanation: There is no common prefix among the input strings.

Q2. Password Checker

You are given a function.

```
int CheckPassword(char str[], int n);
```

The function accepts string str of size n as an argument. Implement the function which returns 1 if given string str is valid password else 0.

str is a valid password if it satisfies the below conditions.

- At least 4 characters
- At least one numeric digit
- At Least one Capital Letter
- Must not have space or slash (/)
- Starting character must not be a number

Assumption: Input string will not be empty.

Example:

Input:

aA1_67

Output:

1

Sample Input:

a987 abC012

Output:

0

Q3. Letter Combinations of a Phone Number

Given a string containing digits from 2-9 inclusive, return all possible letter combinations that the number could represent. Return the answer in any order.

A mapping of digit to letters (just like on the telephone buttons) is given below. Note that 1 does not map to any letters.



Example 1:

Input: digits = "23"

Output: ["ad","ae","af","bd","be","bf","cd","ce","cf"]

Example 2:

Input: digits = ""

Output: []

Example 3:

Input: digits = "2"

Output: ["a","b","c"]

Q4. Java program to find all permutations of a given String using recursion.

For example, given a String "XYZ", this program will print all 6 possible permutations

e.g. XYZ, XZY, YXZ, YZX, ZXY, XYX

Q5. Anagram Check

Given two strings s and t, return true if t is an anagram of s, and false otherwise.

Example 1:

Input: s = "anagram", t = "nagaram"

Output: true

Example 2:

Input: s = "rat", t = "car"

Output: false

Here are some **case study-based programming questions** on **Java Arrays and Strings**, along with **sample input and output**.

Q6. Replace Profanity in Text

Problem Statement:

A social media platform wants to automatically replace **offensive words** with *** in user-generated content. Given a **list of bad words**, replace them in the input text.

Example Input/Output:

Bad Words List: "bad", "ugly", "stupid"

Input:

Enter text: This is a bad and ugly comment.

Output:

Filtered text: This is a *** and *** comment.

Q7. Unique Characters in a String

Problem Statement:

A company is developing a new security system that requires checking whether a given string has all unique characters. You need to write a program that:

- Accepts a string from the user.
- Checks if all characters in the string are unique.

Example Input/Output:

Input 1:

Enter a string: abcde

Output 1:

All characters are unique.

Input 2:

Enter a string: hello

Output 2:

String contains duplicate characters.

Q8. Find the Longest Word

Problem Statement:

A book editor tool requires a function to find the **longest word** in a given sentence.

Example Input/Output:

Input:

Enter sentence: Innovation distinguishes between a leader and a follower.

Output:

Longest Word: distinguishes

Q9. Convert Sentence to Title Case

Problem Statement:

A content editing software needs a function to convert a **sentence** into **title case** (i.e., the first letter of each word should be uppercase).

Example Input/Output:

Input:

Enter sentence: java programming is fun

Output:

Title Case: **Java Programming Is Fun**

Q10. Extract Domain from URL

Problem Statement:

A digital marketing company wants to extract the **domain name** from a given URL. Write a Java program that:

- Accepts a **URL**.
- Extracts and displays only the **domain name**.

Example Input/Output:

Input:

Enter URL: https://www.google.com/search?q=java

Output:

Domain: google.com

*****Note:** You are supposed to use at least two classes and one object for each program.